

AD 502 – Machine Learning

Pre-Reading Material – Lecture 2 The Machine Learning Life Cycle

Estimated Reading Time: 20–25 minutes

Reference Reading

1. Christoph Schröer et al., *A Systematic Literature Review on Applying CRISP-DM Process Model* (2021).
2. D. Sculley et al., *Hidden Technical Debt in Machine Learning Systems* (NeurIPS, 2015).
3. Eric Breck et al., *The ML Test Score: A Rubric for ML Production Readiness* (2017).
4. Nithya Sambasivan et al., *Data Cascades in High-Stakes AI* (CHI 2021).

Why This Topic Matters

Many beginners believe that Machine Learning ends once a model has been trained. In reality, developing a successful Machine Learning application involves a sequence of interconnected stages, beginning with understanding the problem and continuing through data collection, preparation, model development, deployment, monitoring, and continuous improvement.

Industry studies have shown that most effort in Machine Learning projects is spent on understanding data, maintaining deployed systems, and monitoring model performance rather than simply training algorithms. A structured Machine Learning Life Cycle helps organizations develop reliable, maintainable, and scalable AI systems while reducing technical debt and improving long-term performance.

During this lecture, we will study the complete Machine Learning Life Cycle, understand the purpose of each stage, and explore why data quality and continuous monitoring are essential for successful AI systems.

Learning Outcomes

After completing this pre-reading and attending the lecture, you should be able to:

- Explain the stages of the Machine Learning Life Cycle.
- Describe the purpose of each phase in CRISP-DM.
- Distinguish between model development and production deployment.

- Explain why data quality influences every stage of the lifecycle.
- Recognize the importance of monitoring and continuous model improvement.

Activate Your Prior Knowledge

1. How does Netflix continuously improve its movie recommendations?

Think beyond training the recommendation model once. What happens when new users join, movies are added, or user preferences change?

2. Why does Google Maps become more accurate over time?

Millions of users continuously contribute location and traffic information. How might this new data improve route predictions?

3. Suppose a bank develops a fraud detection model.

Will the model continue to perform equally well after one year if fraudsters change their behavior? Why might the model require retraining?

4. Imagine building an AI system for disease diagnosis.

Which would be more harmful:

- using an advanced algorithm with poor-quality data, or
- using a simpler algorithm with high-quality data?

Think about your answer before reading.

Prerequisite Concepts

Before attending this lecture, you should understand:

- Basic definition of Machine Learning
- Difference between traditional programming and Machine Learning
- Training data and testing data



- Importance of learning from data

No knowledge of advanced algorithms is required.

Core Concepts

What is the Machine Learning Life Cycle?

The Machine Learning Life Cycle is a structured process that guides the development, deployment, and maintenance of Machine Learning systems. Instead of viewing Machine Learning as only model training, the lifecycle considers every activity required to build reliable AI applications. The CRISP-DM methodology organizes this process into six iterative phases: Business Understanding, Data Understanding, Data Preparation, Modeling, Evaluation, and Deployment.

Why is the Life Cycle Important?

Successful AI projects depend not only on selecting good algorithms but also on defining the right problem, collecting high-quality data, validating models, deploying them correctly, and continuously monitoring their performance. Real-world ML systems require extensive supporting infrastructure beyond the model itself.

Data is Central to Every Stage

Poor-quality data collected early in the project can create downstream problems throughout the lifecycle. These issues may reduce model accuracy, increase development costs, and even lead to failures after deployment. Researchers describe these compounding effects as *Data Cascades*.

Deployment is Not the End

Unlike traditional software, Machine Learning systems continue to evolve after deployment. New data may change input distributions, user behavior may shift, and model performance can gradually decline. Continuous monitoring, testing, and retraining are therefore essential for maintaining reliable systems.

Production-Ready Machine Learning

A production-ready ML system requires much more than high prediction accuracy. Engineers must ensure reliable data pipelines, reproducible training, robust testing, monitoring, privacy protection, and the ability to update or roll back deployed models when necessary.

Guided Reading Questions



1. Why is model training only one part of the Machine Learning Life Cycle?

Look for examples showing the additional engineering work required before and after training a model.

2. What are the six phases of the CRISP-DM process model?

Identify the objective of each phase and consider how they connect to one another.

3. What is technical debt in Machine Learning?

Observe how poor design decisions today can make future maintenance more difficult.

4. Why is data quality important throughout the lifecycle?

Consider how problems during data collection may affect later stages such as deployment and monitoring.

5. Why must Machine Learning models be continuously monitored after deployment?

Think about changing user behavior, new data, and evolving environments.

Reflection Activity

Write a **150–200 word reflection** describing a Machine Learning application (such as YouTube recommendations, Google Maps, Amazon product suggestions, ChatGPT, or online fraud detection).

Identify the stages of the Machine Learning Life Cycle that you think are involved in building and maintaining the system. Discuss why continuous monitoring and periodic model updates are necessary to keep the application accurate and reliable.

Self-Assessment

Answer the following before class:

1. What is meant by the Machine Learning Life Cycle?
2. List the six phases of the CRISP-DM process model.
3. Why is data quality important throughout the lifecycle?
4. What is technical debt in Machine Learning?

5. Why do deployed Machine Learning models require continuous monitoring and retraining?

